

Md Mehedi Hasan

Boyra, Khulna-9203, Bangladesh

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Academic Credentials

Pabna University of Science and Technology (PUST) | Pabna-6600, Bangladesh [March 2018-April 2024]

Bachelor of Science (B.Sc) in Electrical and Electronic Engineering

CGPA: **3.85 / 4.00** (Ranked **1st** in class of 43 students)

Thesis Title: *Management of EVs for Mitigating Unbalance in a LV Distribution Grid*

Supervisor: *Md. Rabiul Islam, Ph.D*

Khulna University of Engineering & Technology (KUET) | Khulna-6600, Bangladesh [July 2024-present]

Master of Science (M.Sc. Eng.) in Electrical and Electronic Engineering

CGPA: **3.83 / 4.00** Coursework: Completed | **Thesis Proposal:** Approved | **Expected Completion:** 2026

Thesis Title: *A Multi-Level Control Framework for Voltage Quality Improvement in PV-Integrated Distribution Networks*

Supervisor: *Naruttam Kumar Roy, Ph.D*

Research Interest

- Power System Stability.
- Smart Grids.
- Renewable Energy Integration.
- Electric Vehicle
- Microgrid

Teaching Experience

Khulna Khan Bahadur Ahsanullah University | Khulna-6600, Bangladesh [1 september 2024-Present]

Lecturer – Department of Electrical and Electronic Engineering

Faculty Profile: ✉ [Profile](#)

Journal Publications

- **Hasan, M.M.***, Refat, K.A., Ali, M.F. et al. Design and techno-economic validation of a university campus hybrid microgrid using HOMER Pro and DiGSILENT PowerFactory. Sci Rep 16, 17248 (2026). <https://doi.org/10.1038/s41598-026-45872-9> [Link](#)
- Md. Galib Hasan, Md. Farzu Rahman, **Md Mehedi Hasan**, Mahmud Hasan, Md Abbas Uddin, Md. Golam Sadeque, A.H.M. Iftexharul Ferdous, Sustainable charging framework: Solar wireless EV charging enhanced with MPPT control, Energy Reports, Volume 15, 2026, 109274, ISSN 2352-4847, <https://doi.org/10.1016/j.egy.2026.109274>. [Link](#)

Under Review / Submitted

- Swopnil Das, **Md. Mehedi Hasan***, M. Shaharyar Alam, Abu Huraira Taj "A Unified Framework for Hybrid Renewable Energy Microgrid Planning and Operational Feasibility Assessment," International Journal of Energy Research, Wiley (Under Review).
- **M. M. Hasan**, A. H. Shakil, M. R. Islam, and N. K. Roy, "A Hierarchical Control Framework for Voltage Unbalance and Neutral Current Mitigation in PV-Rich Low-Voltage Distribution Networks". (M.Sc thesis), International Journal of Energy Research, Wiley (Submitted).
- M. M. Hasan, A. H. Shakil, N. K. Roy, and M. R. Islam, "Optimal Phase Placement and PV Curtailment for Voltage Unbalance Mitigation in PV-Rich Low-Voltage Networks: A Comparative Study of GA, PSO, and DE". (M.Sc thesis), International Journal of Energy Research, Wiley (Submitted).

Conferences

- **M. M. Hasan**, M. A. Yousuf and M. R. Islam, "Management of EVs for Mitigating Unbalance in a LV Distribution Grid," 2024 6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT), Dhaka, Bangladesh, 2024, pp. 1152-1156, doi: 10.1109/ICEEICT62016.2024.10534474.
- M. R. Rashid, **M. M. Hasan*** and M. R. Islam, "Develop a Method for Mitigating Unbalance Considering EV Users Transportation Needs," 2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN), Rangpur, Bangladesh, 2025, pp. 1-5, doi: 10.1109/QPAIN66474.2025.11171788. [Link](#)
- **M. M. Hasan**, M. R. Islam and M. A. Yousuf, "Mitigating Unbalance Due to EV Dispatch in a LV Distribution Grid Having Balanced Loads," 2024 IEEE International Conference on Power, Electrical, Electronics and Industrial Applications (PEEIACON), Rajshahi, Bangladesh, 2024, pp. 1-4, doi: 10.1109/PEEIACON63629.2024.10800142. [Link](#)
- **M. M. Hasan***, A. H. Shakil, M. R. Islam, and N. K. Roy, "Coordinated Phase Rebalancing of Rooftop PV Systems for Voltage Imbalance Mitigation in LV Networks," International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure 2026 (PECCII), PUST, Pabna, Bangladesh, Jun. 17-18, 2026 (Presented).
- M. W. Ali, **M. M. Hasan***, M. A. Yousuf, and M. I. Ahmad, "Optimal Capacitor Placement and Sizing in Unbalanced Distribution Systems Using a Chaotic Iterative Local Search with Adaptive Descent Algorithm, International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure 2026 (PECCII), PUST, Pabna, Bangladesh, Jun. 17-18, 2026

(Presented).

- R. Islam, M. A. M. Jesun, M. A. Yousuf, **M. M. Hasan***, and M. S. Alam, "Optimization and Feasibility Analysis of a Grid-Connected Hybrid Renewable Microgrid for Cost and Emission Reduction in Bangladesh," International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure 2026 (PECCII), PUST, Pabna, Bangladesh, Jun. 17-18, 2026 (Presented).
- P. Chowdhury, S. Das, M. A. Yousuf, **M. M. Hasan***, and Md. F. Ali, "Techno-Economic Optimization of a Grid-Connected PV-Wind-BESS Microgrid for Commercial Loads in Bangladesh," International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure 2026 (PECCII), PUST, Pabna, Bangladesh, Jun. 17-18, 2026 (Presented).

Skills & Expertise

- **Programming Language:** C, C++, MATLAB and Python.
- **Engineering Design & Simulation:** Microsoft Visio, Edraw Max, PVsyst, CircuitLab, Simulink.
- **Productivity & Office Tools:** Microsoft Excel, PowerPoint, Word.

Projects

- A project on Rice Leaf Disease Detection by using Transfer Learning. [Link](#) [March 2024-May 2024]
- A project on the Recent Dengue Outbreak in Bangladesh: Prediction and Analysis. [Link](#) [April 2024-June 2024]
- A project on Solar energy forecasting by using ML. [Link](#) [Sep 2024-Present]
- A project on Alzheimer's Disease Detection in MRI images by Deep Learning. [Link](#) [Nov 2023-Jan 2024]

Internship

Intern, Rajbari Palli Bidyut Samity | Rajbari, Bangladesh

[April 2023-June 2023]

Gained hands-on experience in rural 11 kV/0.4 kV distribution systems, including transformer maintenance, line inspection, and consumer service operations.

Awards & Certificates

- Received a certificate for completing the online course Machine Learning Specialization, Advanced and Unsupervised Learning, Recommenders, Reinforcement Learning on Coursera. [Sep 2024-Nov 2024]
- Academic merit scholarship for a Bachelor of Science (B.Sc) in Engineering. [Nov 2019-Apr 2024]

Extracurricular Activities

- Active Center for Research Innovation & Development (CRID) member, contributing to research projects in AI, Image Processing, and EEG Analysis. [2023-Present]
- Organizing Secretary at Jonaki (One of the leading voluntary organizations at PUST, which works for the welfare of students, including raising funds, planting trees, and running free blood donation campaigns). [2022-2023]
- Science Club Coordinator, Shahid Bulbul Govt. College, Pabna. [2016-2017]

Professional Affiliation

- Associate Member, IEEE, Pabna University of Science & Technology, Student Branch, Bangladesh.

References

Md. Rabiul Islam

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Naruttam Kumar Roy

Professor

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